

ERNIE Toolkit Build of Materials and Parts Checklist

Carrying Case

No.	Part	Part Number	Notes	Quantity	Total Cost(Canadian dollars)	3D Printed (Yes/No)	On-site for Assembly ✓ (Yes/No) ☐
1.	Case	AL2 1769 009	The Nanuk 935 hard case can also be used	1	\$90.55	No	

Subtotal: **\$90.55**

Magnet

No.	Part	Part Number	Notes	Quantity	Total Cost (Canadian dollars)	3D Printed (Yes/No)	Added to Assembly ✓ / Not yet added to the Assembly ☐	Disassembled ✓ / Not yet Disassembled ☐
1.	Ring 42 (with PLA)*		Approximately(Appr.) 134g of PLA per ring	2	-	Yes		
2.	Ring 251(with PLA)*		Appr. 134g of PLA per ring	2	-	Yes		
3.	Ring 288		Appr. 134g of	2	-	Yes		

No.	Part	Part Number	Notes	Quantity	Total Cost (Canadian dollars)	3D Printed (Yes/No)	Added to Assembly ✓/ Not yet added to the Assembly ☐	Disassembled ✓/ Not yet Disassembled ☐
	(with PLA)*		PLA per ring					
4.	Ring 321 (with PLA)*		Apr. 134g of PLA per ring	1	-	Yes		
5.	Lids for Rings 42 (with PLA)*		Apr. 45g required to print each lid	2	-	Yes		
6.	Lids for Rings 251		Apr. 45g required to print each lid	2	-	Yes		
7.	Lids for Rings 288 (with PLA)*		Apr. 45g required to print each lid	2	-	Yes		
8.	Lids for		Apr. 45g required to	1	-	Yes		

No.	Part	Part Number	Notes	Quantity	Total Cost (Canadian dollars)	3D Printed (Yes/No)	Added to Assembly ✓ / Not yet added to the Assembly ☐	Disassembled ✓ / Not yet Disassembled ☐
	Ring 321 (with PLA)*		print each lid					
9.	180mm Brass Rods	90162A050	Two 1m brass rods can be bought. 1 brass rod = five 160mm rods	8(166mm each)	\$39.54	No		
10.	Ring Spacers (3D printed)		Appr. 100g of PLA is required to print 80 ring spacers	48	-	Yes		
11.	M4 Brass Nuts	B0CQGYC7JZ	50 pieces	8	\$11.33	No		
12.	M3 Nylon /	a23122400ux0074	Brass screws are preferred-	54	\$13.99	No		

No.	Part	Part Number	Notes	Quantity	Total Cost (Canadian dollars)	3D Printed (Yes/No)	Added to Assembly ✓/ Not yet added to the Assembly ☐	Disassembled ✓/ Not yet Disassembled ☐
	Brass Screws		60 pieces per packet					
13.	Dome-Shaped Nuts	90527A505	100 pieces per packet	8	\$22.96	No		
14.	Washers	95610A150	100 pieces per packet	16	\$7.70	No		
15.	Neodymium Magnets (12mm X12mm X12mm)		Specify dimensions to the vendor. \$5.87 per magnet	98	\$575.26	No		
16.	Reference Neodymium Magnets		These are made from the set of Neodymium magnets purchased	4	\$23.48	No		
17.	M3 Brass Heat	a24062300ux0395	100 pieces per packet	54	\$17.69	No		

No.	Part	Part Number	Notes	Quantity	Total Cost (Canadian dollars)	3D Printed (Yes/No)	Added to Assembly ✓/ Not yet added to the Assembly ☐	Disassembled ✓/ Not yet Disassembled ☐
	Set inserts							
18.	PLA , PETG, ABS		Different colors depending on choice.	4	\$25.99	No		

Subtotal: **\$737.94**

No.	Part	Notes	Quantity	Total Cost (Canadian dollars)	3D Printed (Yes/No)	On-site for Assembly ✓ (Yes/No) ☐
1.	Neodymium magnets	These are necessary to replace broken magnets	10	\$58.7	No	

Subtotal: **\$58.7**

Tools needed for Magnet Assembly

No.	Tool	Part number	Notes	Quantity	Total cost	Available (Yes/No)	Onsite for Assembly ✓/ ✓/	Returned to box ✓/

							Not onsite for Assembly ☐	Not returned to Box ☐
1.	Screw Drivers Screw Drivers	06973932 450615	Select a crosshead / philips shaped with a tight grip	2 2	\$20.99 \$33.98	No		
2.	Spanners	hta25010 6hh00121 5	7mm Open Wrench	2	\$25.59	No		
3.	Compass			2	\$54.99	No		
4.	Permanen t Markers	SAN3511 3PP		4	\$13.78	No		
5.	Vernier Caliper	USQC033 01915	6 inch	1	\$34.99	No		
6.	Ruler	4	Must be non-magnetic	1	\$6.69	No		

Subtotal: **\$191.01**

Robot Assembly

No.	Part	Part Number	Part Number	Quantity	Notes	Total Cost	Available (Yes/No)	Added to Assembly ✓/ Not yet added to	Disassembled ✓/
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						(Canadian dollars)		the Assembly ☐	Not yet Disassembled ☐
1.	Magnet Rails (1 set of 3)	T06		3	(Please note that all three axes, X, Y, Z should be 150mm)	\$273.78	No ▾		
2.	Motor Drivers			3		\$34.62	No ▾		
3.	Arduino Uno	ABX00080		1		\$33.50	No ▾		
4.	3D Printed Bases (PLA)			4			No ▾		
5.	Gauss Meter	GM2		1		\$1200	No ▾		
6.	Screws 4-M4 and 4-M3			8	These come along with the rails.		No ▾		
7.	Connecting Wires	22awg		25	Cut out specific lengths	\$23.99	No ▾		

8.	Power Supply	CA-24V 3A-72W		1		\$22.99	No ▾		
9.	3D Printed Gauss Meter Probe Holder (PLA)			1			No ▾		
10.	Laptop	HP 15		1	(Install Ubuntu to use MaRGE)	\$689.99	No ▾		
11.	Gauss meter probe calibrator (add to Gauss meter)	ZGC		1		\$196	No ▾		
12.	Gauss Meter Casing	HC-GM2		1	Add to the Gauss meter purchase	\$33	No ▾		
13.	Gauss Meter batteries	DRY019 6		Set of 12		\$35.43	No ▾		

	3D printed Motor driver casing (PLA)						No ▾		
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Subtotal: **\$2543.3**

Tools needed for Robot Assembly

No.	Tool	Notes	Quantity	Total Cost	Available (Yes/No)	Returned to box ✓ / Not returned to the box ☐
1.	Screw Drivers	One can buy a set of screw drivers; however, flat head screw drivers are preferred.)	2	N/A*	No ▾	
2.	Allen Keys	Hex Bit Set	4	\$25.98	No ▾	
3.	Masking Tape	AB506-3	1	\$9.05	No ▾	

Subtotal: **\$56.02**

**Previously captured*

Magnet Disassembly

No.	Part	Part Number	Quantity	Notes	Total cost	Available	On-site for Assembly ✓ (Yes/NO)
1.	Magnet Ouster	871392000899	1	Sharpen them to push out	\$11.54	No ▾	

				magnets from rings easily			
2.	3D Printed Disassembly Tray (PLA)		1			No ▾	
3.	3D Printed Disassembly Tray Cover (PLA)		1			No ▾	
4.	Cutter (to sharpen the Ousters)	016392515244	1		\$8.99	No ▾	
5.	Brass Heat set inserts	a24062300ux0395	9	M3- same as the heat set inserts used in the rings		No ▾	
6.	Brass Screws- non magnetic		9	Same as brass screws used with rings		No ▾	

Subtotal: **\$20.53**

Electronics required for signal generation

No.	Part	Part Number	Quantity	Notes	Total Cost(USD) (1USD = 1.4Canadian Dollar)	Available (Yes/No)	On-site for Assembly ✓ (Yes/No) ☐
1.	Red pitaya	IZD0021	1		\$1234.65	No ▾	
2.	RF power amplifier (RFPA)	ZX60-100V H+	1		\$476.28	No ▾	
3.	RF Pre-amplifier (LNA) (ABL 0050-00-4510)	ABL0050-00-4510	1	Or Narrow BW NMR pre-amplifier	\$397.60	No ▾	
4.	RFPA power supply	2260-KD300 5D	1	Supply 12V and 300mA	\$138.04	No ▾	
5.	LNA power supply	2260-KD300 5D	1	Supply 12V and 100mA	\$138.04	No ▾	
6.	Passive TR switch		1	PCB printed and electronics purchased following the OSI² switch part list	\$100	No ▾	
7.	RF Coil (3D printed)		1			No ▾	

8.	Copper enamelled wire for winding RF coils		1		\$211.28	No ▾	
9.	RF coil holder (3D printed)		1			No ▾	
10.	Variable tuning and matching capacitors (for RF matching)	PPZN30100	2	Wide range variable capacitors to be added to fixed capacitors	\$156.94	No ▾	
11.	Phantom (3D printed)		1	3D print cylindrical part and fill it with olive oil		No ▾	
12.	SMA cables	CSE-SGAM-305-SGAM	12	Purchase like 20+	\$146.76	No ▾	
13.	SMA adaptors-F-F	RFAD-SA3S A2-01-N-A	8		\$43.76	No ▾	
14.	Aluminum box for LNA	CU-234	1		\$19.33	No ▾	
15.	Aluminum box for RFPA	CU-234	1		\$19.33	No ▾	

16.	Aluminum box for red pitaya and FHDO	1590Z166	1		\$223.13	No ▾	
17.	Aluminum box for passive TR switch	CU-234	1		\$19.33	No ▾	
18.	Aluminum box for Matching Circuit	CU-234	1		\$19.33	No ▾	
19.	Aluminum sheet for scanner shielding	BSS-00903	1	Cut this out to fit the size of the scanner. Drill holes where the dome-shaped nuts will pass	\$43.30	No ▾	
20.	Copper sheet for RF shielding	60	1	Cut this out to fit inside the magnet. Solder an earthing wire on the end and connect to the aluminium shield and brass rod to establish	\$30.99	No ▾	

				common ground.			
21.	Copper tape	B083QJDQ5 M	1	Used to hold together the copper shield	\$12.59	No ▾	
22.	Aluminum tape	143650	1		\$39.80	No ▾	
23.	Brass screws	48-133	9		\$36.63	No ▾	
24.	Ground cables	1GY89.500 1B	5		\$28.31	No ▾	
25.	SMA Female to BNC Male Adapter	ADP-SMAF-BNCM	4		\$33.64	No ▾	
26.	BNC cables 15 cm (for VNA testing)	115101-19-0 6.00	4		\$70.80	No ▾	
27.	Test lead banana to grabber 36'	3782-36-02	6		\$201.72	No ▾	
28.	Olive oil for phantom		1	Any olive oil from a grocery store could work.	\$16.79	No ▾	

29.	Ethernet cable	N201P-006-BL	2		\$34.62	No ▾	
30.	Masking tape	3436-3	1		\$11.05	No ▾	

Subtotal: **\$3904.04**

Required Lab equipment

No.	Part	Quantity	Total Cost	Available	On-site for Assembly ✓ (Yes/NO) ☐
1	Label maker	1	\$64.99	No ▾	
2	Drilling machine	1	\$327.94	No ▾	
3	Soldering gun	1	\$258	No ▾	
4	Nano VNA	1	\$79.99	No ▾	
5	Function generator	1	\$379	No ▾	
6	Multi meter	1	\$227.73	No ▾	
7	Oscilloscope	1	\$637.43	No ▾	
8	VNA-SVA1015 X	1	\$1944.00	No ▾	
9	cutter	1	\$70	No ▾	

Subtotal: **\$2045.08**

Other Important Lab Equipment

No.	Equipment	Part Number	Quantity	Total Cost	Available (Yes/No)
1.	3D Printer	737896686575063052	1	\$739	No ▾
2.	PLA Spools (for rings, lids, spacers, and other printed parts)	A00-G6-1.75-1000-SPL	3 kg (5 spools) 4 green, 1 white	\$129.95	No ▾

Subtotal: \$868.95

Other Important Electronics

No.	Electronics	Quantity	Notes	Total Cost	Available (Yes/No)
1.	Pickup coil	1	Made from BNC cables.	-	No ▾
2.	Set of Ceramic Capacitors (1pF- 800pF)	50>	Different values of ceramic fixed capacitors to be used during tuning and matching. (Purchase a	\$100	No ▾

No.	Electronics	Quantity	Notes	Total Cost	Available (Yes/No)
1.	Pickup coil	1	Made from BNC cables.	-	No ▾
			certain quantity depending on your budget)		

Grand Total Estimate: \$10516.12